

TrES-3b

R-1563

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190324 · created 2026-08-01T21:46:35+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458566.8576 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.011
Transit depth (Rp/Rs)^2	3.27 +/- 0.38 [%]
Orbital Inclination (inc)	85.9 +/- 0.81
Airmass coefficient 1 (a1)	0.972 +/- 0.015
Airmass coefficient 2 (a2)	0.0141 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#1 - [502, 185]
Optimal Method	PSF photometry
Transit Duration (day)	0.0794 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R _★ fit vs published	0.181 ± 0.011 vs 0.16309 (deviation 1.3σ)
Duration fit vs published	0.0794 d vs 0.0567 d (deviation 6.5σ)
Mid-time O–C	-4.1 min
Depth SNR	13.2
Residual scatter	1.53 %

Light curve

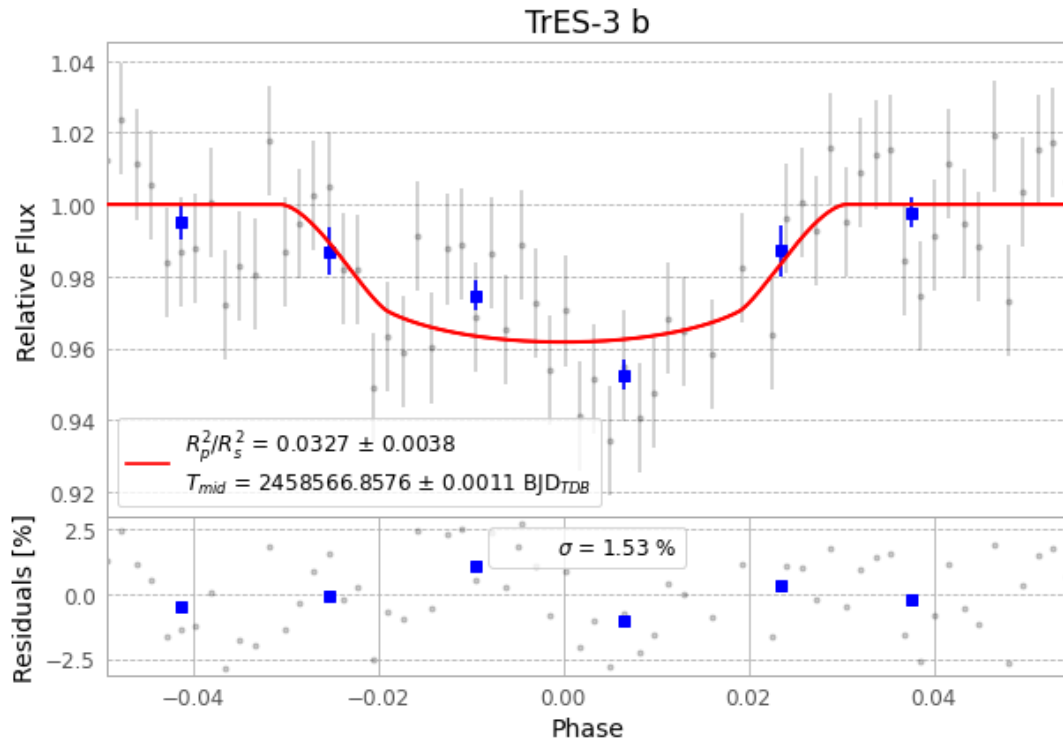


Figure 1 — FinalLightCurve_TrES-3 b_2019-03-24.png (SHA-256 709e6f1e2ddd78c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 28], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9fa5fc5c0daeddd3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3190324070012.FITS → TRES-3190324101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190324.log (9ea552a24ebf...), FinalLightCurve_TrES-3 b_2019-03-24.png (709e6f1e2ddd...), FinalLightCurve_TrES-3 b_2019-03-24.csv (197e2afab7cd...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:46Z.